ANATOMY · MCQ PRACTICE

Urinary Tract & Posterior Abdominal Wall — MCQ Practice

SGL 1 · Dr. Faris Muhammed
GUE BLOCK

Anatomy GUE Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.
4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. What is the approximate length of the ureter given in the lecture? EASY

A. 10–12 cm
B. 25–30 cm
C. 4–6 cm
D. 14–16 cm
E. 40–45 cm

2. Which layer of the suprarenal cortex produces aldosterone? EASY

A. Zona fasciculata
B. Zona reticularis
C. Medulla
D. Zona glomerulosa
E. Capsule

3. Which covering of the kidney lies immediately outside the renal capsule? EASY

A. Perirenal fat
B. Renal (Gerota's) fascia
C. Pararenal fat
D. Transversalis fascia
E. Parietal peritoneum

4. Which nerve supplies the external urethral sphincter? EASY

A. Pelvic splanchnic nerves
B. Greater thoracic splanchnic nerve
C. Pudendal nerve (S2–S4)
D. Hypogastric plexus (T11–L2)
E. Obturator nerve

5. During dissection a student follows a slender muscle from the bodies of T12 and L1 down along psoas major. Where does it insert? MEDIUM

A. Lesser trochanter, through the iliopsoas tendon
B. Greater trochanter of the femur
C. Lower border of the 12th rib
D. Posterior part of the iliac crest
E. Pectineal line and iliopubic eminence

6. The lateral arcuate ligament of the diaphragm arches over which muscle? MEDIUM

A. Psoas major
B. Quadratus lumborum
C. Iliacus
D. Transversus abdominis
E. Psoas minor

7. Which nerve pierces psoas major and then runs down on its anterior surface? MEDIUM

A. Genitofemoral nerve (L1, L2)
B. Femoral nerve (L2–L4)
C. Obturator nerve (L2–L4)
D. Ilio-inguinal nerve (L1)
E. Lateral femoral cutaneous nerve

8. Which nerve of the posterior abdominal wall crosses iliacus and leaves by passing under or through the lateral part of the inguinal ligament? MEDIUM

A. Femoral nerve (posterior divisions L2–L4)
B. Genitofemoral nerve (L1, L2)
C. Iliohypogastric nerve (L1)
D. Lateral femoral cutaneous nerve
E. Obturator nerve (anterior divisions)

9. A surgeon exposing the front of the RIGHT kidney finds a retroperitoneal organ lying over its medial part near the hilum. Which organ is it? MEDIUM

A. Jejunum
B. Stomach
C. Duodenum
D. Spleen
E. Sigmoid colon

10. During a left adrenalectomy the surgeon ligates the left suprarenal vein. Into which vessel does it normally drain? MEDIUM

A. Inferior vena cava directly
B. Left gonadal vein
C. Splenic vein
D. Portal vein
E. Left renal vein

11. A drug that mimics the parasympathetic (S2–S4) supply to the bladder would have which effect? MEDIUM

A. Detrusor relaxes and the internal sphincter contracts
B. Detrusor contracts and the internal sphincter relaxes
C. Only the external sphincter contracts
D. Detrusor and internal sphincter both contract
E. Detrusor and internal sphincter both relax

12. A man has loin-to-groin colic. CT shows a stone stuck where the ureter crosses the bifurcation of the common iliac artery. Which natural narrowing is this? MEDIUM

A. Crossing of the pelvic brim
B. Pelviureteric junction
C. Entry into the bladder wall
D. Junction of minor and major calyces
E. Intermediate (membranous) urethra

13. During a hysterectomy the uterine artery is clamped beside the cervix near the lateral fornix. Which structure passes just beneath the artery here and is at risk? MEDIUM

A. Obturator nerve
B. Round ligament
C. Ovarian artery
D. Ureter
E. Internal iliac vein

14. The ejaculatory ducts open on the seminal colliculus. In which part of the urethra is this? MEDIUM

A. Intermediate (membranous) urethra
B. Bulbous part of the spongy urethra
C. Prostatic urethra
D. Penile (pendulous) urethra
E. Navicular fossa

15. Why is the suprarenal medulla described as behaving like a sympathetic ganglion? HARD

A. Postganglionic fibres from the coeliac ganglion end on its cells
B. Pelvic splanchnic fibres from S2–S4 synapse inside the medulla
C. Vagal fibres relay in the medulla on their way to the gut
D. Branches of the pudendal nerve supply the medulla directly
E. Greater splanchnic preganglionic fibres end directly on its cells

16. Which statement correctly contrasts the ribs lying behind the two kidneys? HARD

A. Both kidneys: 11th and 12th ribs
B. Left kidney: 11th and 12th ribs; right kidney: 12th rib only
C. Right kidney: 11th and 12th ribs; left kidney: 12th rib only
D. Both kidneys: 12th rib only
E. Left kidney: 10th–12th ribs; right kidney: 11th rib only

17. Which structure lies in front of the LEFT ureter but is not an anterior relation of the right ureter? HARD

A. Sigmoid mesentery
B. Duodenum
C. Ileal mesentery
D. Genitofemoral nerve
E. Psoas major

18. After a fall astride a beam, a man has blood and urine in the scrotum, penis and lower anterior abdominal wall. Which part of the urethra is torn? HARD

A. Prostatic urethra
B. Intermediate (membranous) urethra
C. Spongy (bulbar) urethra
D. Internal urethral orifice
E. Pelviureteric junction

19. A pelvic fracture tears the intermediate (membranous) urethra. Where does the extravasated blood and urine collect? HARD

A. Scrotum and penis, deep to Colles' and Scarpa's fascia
B. Within the peritoneal cavity, among the bowel loops
C. In the ischiorectal fossa beside the anal canal
D. Down into the thigh along the femoral sheath
E. Above the perineal membrane, around the prostate and bladder

20. What is the correct order of arterial branching inside the kidney? HARD

A. Interlobar → segmental → arcuate → interlobular
B. Segmental → arcuate → interlobar → interlobular
C. Arcuate → interlobar → segmental → interlobular
D. Segmental → interlobar → arcuate → interlobular
E. Segmental → interlobular → arcuate → interlobar

✓ Answer key

Q	Answer	Difficulty	Why
1	**B**	EASY	The ureter is 25–30 cm long, running from the renal pelvis to the posterior wall of the bladder.
2	**D**	EASY	Zona glomerulosa → mineralocorticoids (aldosterone); fasciculata → cortisol; reticularis → androgens; medulla → adrenaline.
3	**A**	EASY	Inner → outer: renal capsule → perirenal fat → renal (Gerota's) fascia → pararenal fat.
4	**C**	EASY	The pudendal nerve is the somatic supply (+) to the external urethral sphincter; the autonomic nerves act on the detrusor and internal sphincter.
5	**E**	MEDIUM	Psoas minor: bodies of T12–L1 and the disc → pectineal line and iliopubic eminence through the iliac fascia. Psoas major and iliacus go to the lesser trochanter.
6	**B**	MEDIUM	Lateral arcuate → quadratus lumborum; medial arcuate → psoas major; median arcuate → aorta.
7	**A**	MEDIUM	Genitofemoral is the only nerve on the front of psoas; femoral leaves its lateral border, obturator its medial border.
8	**D**	MEDIUM	The lecture figure shows the lateral femoral cutaneous nerve passing under or through the lateral margin of the inguinal ligament.
9	**C**	MEDIUM	Right kidney, anterior: liver (peritoneum-covered) plus retroperitoneal adrenal, bare area of liver, duodenum and colon. Jejunum, stomach and spleen relate to the left kidney.
10	**E**	MEDIUM	Left adrenal vein → left renal vein; right adrenal vein → IVC directly (lecture figure).
11	**B**	MEDIUM	Pelvic splanchnic (S2–S4): detrusor +, internal sphincter − → voiding. Sympathetic T11–L2 does the reverse.
12	**A**	MEDIUM	The second ureteric constriction is at the pelvic brim, where the ureter lies on the common iliac bifurcation; the other two are the pelviureteric junction and bladder entry.
13	**D**	MEDIUM	In the pelvis the ureter passes under the uterine artery and is related to the lateral fornix (relations slide + uterine artery figure).
14	**C**	MEDIUM	The urethral crest of the prostatic urethra carries the seminal colliculus, with the prostatic utricle and the ejaculatory duct openings.
15	**E**	HARD	Preganglionic splanchnic fibres ramify around medulla cells without synapsing; the coeliac ganglion's postganglionic fibres only supply the gland's vessels.
16	**B**	HARD	Posterior relations figure: the left kidney is crossed by the 11th and 12th ribs, the right by the 12th only.
17	**A**	HARD	Left ureter: left gonadal a., left colic a., sigmoid mesentery in front. Duodenum and ileal mesentery are right-sided; psoas and genitofemoral lie behind both.
18	**C**	HARD	Anterior (bulbar/penile) injury is extra-abdominal: extravasation runs deep to Colles' and Scarpa's fascia into scrotum, penis and abdominal wall.
19	**E**	HARD	Posterior (membranous/prostatic) injury: extravasation stays above the perineal membrane and extraperitoneal (slide: "intra-abdominal / extraperitoneal").
20	**D**	HARD	Renal a. → segmental (apical, anterior superior, anterior inferior, inferior, posterior) → interlobar → arcuate → interlobular.